Document title

Subtitle of document

GUARDANT™

Conquering Cancer With Data

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GUARDANT™

Conquering Cancer With Data

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1 Introduction

1.1 YAML header

Configure the YAML header including the following elements:

- `title`: Title
- `subtitle`: Subtitle; remove option completely if you don't need a subtitle.
- `author`: Here you can list various author's names including their affiliation(s) and the email of the corresponding author.
- `date`: A date
- `abstract`: The abstract will be shown right after the title in smaller font size.
- `logo1` and `logo2`: path of the logos that will be displayed on the bottom left and right of title page (after the front cover).
- `language`: If language is NOT English refer to th file `custom_lang.yml` for automatic text in German. If the document language is another one, adjust this file.
- `bibliography`: A path to the bibliography file to use for references (BibTeX `.bib` file). The current file includes 3 dummy references; either insert your references into this file or replace the file with your own.
- `csl`: The style is provided in the 'sage-harvard.csl' file, which adopts the SAGE Harvard reference style. Just leave the file as it is.
- `format - pdf`: In this template many of the Quarto options for Word output are listed in the YAML header. If you want to know more about these settings I recommend the PDF format reference for a complete list of available options. For instance, you can adjust the figure and table references with `fig-title`, `tbl-title`, `fig-prefix`, and `tbl-prefix`.
    - `template-partials`: these 3 files are required for the layout of the front cover and title page.
    - `include-in-header`: The file 'styles/in-header.tex' determines the overall layout of the document.
    - Custom options:
        - `cover-bg-image`: path of the image on the cover page.
        - `cover-page-color`: background color of the cover page (provided as hex color code).
        - `cover-text-color`: color of the header on the cover page (provided as hex color code).
        - `cover-fade-effect`: if 'true' (default), the background color will overlay

the image and fade away from top to bottom. If 'false' the image will overlay the background color.

- `execute`: Global options for customizing output from executed code are specified within this execute key.
- `fig-align`: Global settings for figure positioning. For other settings see the PDF format reference.

The default font for this template is 'Helvetica' but can be replaced with the Guardant Health's own font style 'HelveticaNeue'.You can choose the font in the template when running the function, e.g., `GHformats::create_quarto_doc(dirname = "your_pdf_report, template = "pdf_report", font = "HelveticaNeue").`

1.2 Code blocks

Code blocks in Quarto documents are treated in similar way as in Markdown documents. One important difference is that code chunk options (in Quarto also called 'cell level options') are typically included in special comments using `#|` at the top of code chunks rather than within the line that begins the chunk:

Please note that individual words are separated with a hyphen, not a dot, followed by a colon, not an equal sign as in R Markdown documents. Quarto uses this approach to both better accommodate longer options like `fig-cap`, `fig-subcap`, and `fig-alt` as well as to make it straightforward to edit chunk options within more structured editors that don't have an easy way to edit chunk metadata (e.g. most traditional notebook UIs).

However, if you prefer it is still possible to include chunk options on the first line (e.g. "'{r, echo = FALSE}) as in R Markdown documents.

1.3 Callout blocks

Quarto provides five different types of callouts that are an excellent way to draw extra attention to certain concepts.

> **Note**
>
> The color and icon will be different depending upon the type that you select. You can choose between: `note`, `warning`, `important`, `tip`, and `caution`.

Tip With Caption

This is an example of a callout with a caption.

2 Methods

2.1 Cross-references

External images and R figures can be referenced with `@fig-label`, where 'label' is the name of the code chunk. These label names should not contain underscores to separate words, use hyphens here instead. Note that figures need to have a caption to be numbered and for cross-referencing, The caption is also set in the chunk option with `#| fig-cap: "Your caption"`.

Tables require similarly a label and table caption for cross-referencing. But here, the cross-reference contains the prefix 'tbl': `@tbl-label`.

Cross-references to individual sections can simply be made with the prefix 'sec' and by adding a '{#sec-identifier}' to any heading.

This is for example a cross-reference to Table 3.2 in Section 3.2 and a cross-reference to Fig. 3.1 in Section 3.3.

To create a reference-able code block, add a #lst-identifier along with a lst-cap attribute inside the curly braces (see code chunk example Listing 2.1). Note that the indication of the programming language requires for this static code block a dot set before the 'r'.

Listing 2.1 Example for a referenceable code block

```
4+4
```

2.2 Mathematical equations

Use mathematics as usual with the dollar sign `$` at the beginning and end of the equation; either in **inline mode** with one dollar sign such as $E = mc^2$ or in **display mode** with two dollar signs:

$$E = mc^2$$

Important to note: do not leave a space between the '$' and your mathematical notation.

Alternatively, you can use LaTeX for more control and when equations are more complicated. LaTeX equations are also automatically numbered if you define a label within the equation environment, which is useful if you have many equations and want to cross-reference them. The equation label needs to be written with '#eq:label' before the end of the equation (see Equation 2.1):

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \tag{2.1}$$

Formulas and corresponding explanations should be integrated into the sentence and, thus, end with a comma or period. Here comes an example:

If the random variable Y follows a standard normal distribution, i.e. $Y \sim N(0,1)$, it's density function can be described with

$$f_Y(y) = \varphi(y) \stackrel{\text{def}}{=} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\}, \quad y \in \mathbb{R}. \tag{2.2}$$

π represents the circle number or Ludolph's number. The function

$$F_Y(y) = \Phi(y) \stackrel{\text{def}}{=} \int_{-\infty}^{y} \varphi(x)\,\mathrm{d}x, \quad y \in \mathbb{R} \tag{2.3}$$

represents then the distribution function of Equation 2.2.

The numbering of equations, as in Equation 2.2, should only be done if they are referred to in the rest of the text. Especially if there are many equations in the thesis, the use of LaTeX seems to make more sense.

2.3 Images

Quarto includes several features aimed at making it easier to work with figures and subfigures, as well as for laying out panels that contain multiple figures, tables, or other content.

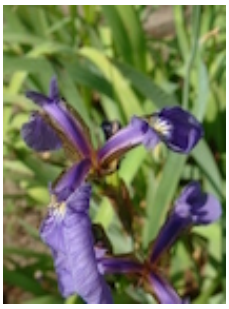
Figure 2.1: Single image of Iris setosa with URL link but no cross-reference.

For instance, if you have several figures that appear as a group, you can create a figure div to enclose them (see Fig. 2.2 and Fig. 2.3).

Figure 2.2: Iris versicolor

Figure 2.3: Iris virginica

The layout attribute enables the creation of much more complex layouts. Fig. 2.4 provides an example with a common figure caption and one identifier for all three.



(a) Iris setosa

(b) Iris versicolor

(c) Iris virginica

Figure 2.4: Custom layout of images

3 Results

3.1 R output

R output is typically shown in the monospace font (here an example with the `mtcars` dataset in the subfolder `data/`):

```
      mpg             cyl             disp             hp      
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5  
 Median :19.20   Median :6.000   Median :196.3   Median :123.0  
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7  
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0  
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0  
```

3.2 Tables

Here is a simple table based on Markdown Syntax (Table 3.1).

Table 3.1: My Caption

A	New	Table
left-aligned	center-aligned	right-aligned
$123	$456	$789
italics	~~strikethrough~~	**boldface**

3.2.1 Using the knitr package

Table 3.2 is an example of using `knitr::kable()` to generate the table.

Table 3.2: This is a table produced with knitr::kable().

	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21.0	6	160	110	3.90	2.620

Table 3.2: This is a table produced with knitr::kable().

	mpg	cyl	disp	hp	drat	wt
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875
Datsun 710	22.8	4	108	93	3.85	2.320
Hornet 4 Drive	21.4	6	258	110	3.08	3.215
Hornet Sportabout	18.7	8	360	175	3.15	3.440

3.2.2 The xtable package

Another useful package for tables for PDF output is xtable. The following code will produce an example table if the *xtable* package is installed. Note that you need to add the chunk option `results = "asis"` inside `{r}` otherwise the PDF will contain the LaTeXcode of the table!

3.3 Figures

A base graphics scatterplot (Fig. 3.1).

Here for comparison a boxplot with a different image height (Fig. 3.2).

Table 3.3: A table made with xtable.

	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21.00	6	160.00	110	3.90	2.62
Mazda RX4 Wag	21.00	6	160.00	110	3.90	2.88
Datsun 710	22.80	4	108.00	93	3.85	2.32
Hornet 4 Drive	21.40	6	258.00	110	3.08	3.21
Hornet Sportabout	18.70	8	360.00	175	3.15	3.44

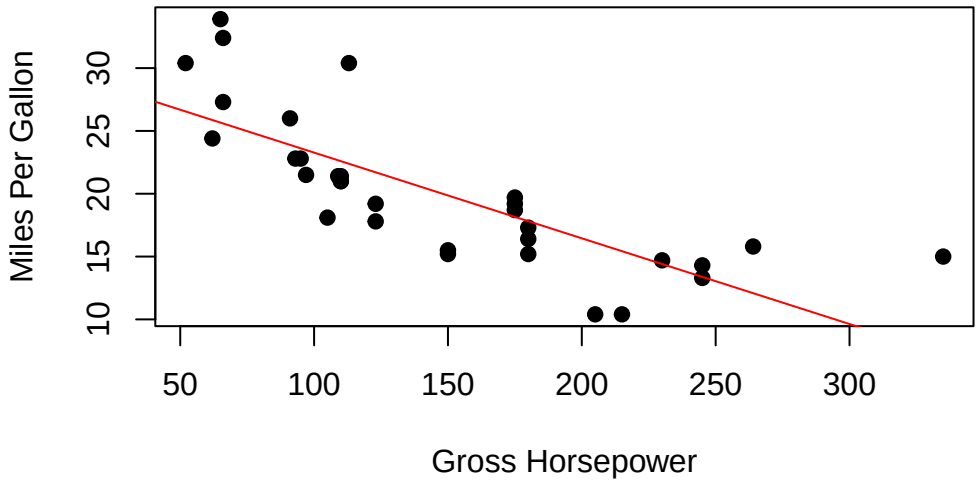


Figure 3.1: Relationship between horsepower and fuel economy.

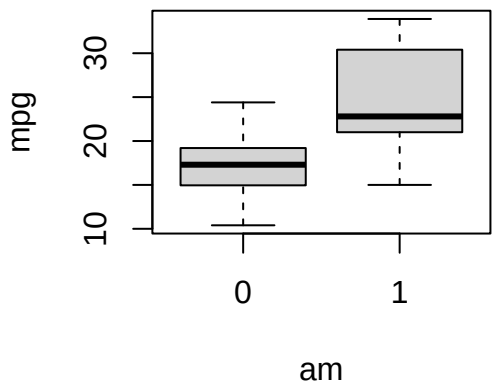


Figure 3.2: Fuel differences between transmission types (0 = automatic, 1 = manual).

4 Adding citations and bibliography

Link a `.bib` document via the YAML header, and the bibliography will be printed at the end. The default bibliography style is provided in the `sage-harvard.csl` file (do not delete), which adopts the SAGE Harvard reference style.

References can be cited directly within the document using the Quarto equivalent of the LaTeXcitation system `[@key]`, where key is the citation key in the first line of the entry in the .csl file. Example: (Taylor and Green, 1937). To cite multiple entries, separate the keys by semicolons (e.g., (Kamm, 2000; Knupp, 1999). You can also write in-text citations, as follows: Taylor and Green (1937) say this and that.

There is also the package citr which I highly recommend: *citr* provides functions and an RStudio add-in to search a BibTeX-file to create and insert formatted Markdown citations into the current document. If you are using the reference manager Zotero the add-in can access your reference database directly.

4.1 Software

If you want to include a paragraph on the software used, here is some example text/code to get the current R and package versions. The code to create a separate bibliography file named 'packages.bib' with all package references has already been added at the beginning of this script (code chunk 'generate-package-refs').

All analyses were performed using the statistical software R (version 4.5.1) (R Core Team, 2025). This thesis, including tables, was generated using the R packages 'rmarkdown' (version 2.30) (Allaire et al., 2025), 'GHformats' (version 1.0.0.9000) (Hintz and Otto, 2022), and 'knitr' (version 1.51) (Xie, 2025).

4.2 Examples for footnotes

Here is a footnote reference[1], and another[2].

[1] Here is the footnote.

[2] Here's one with multiple blocks.

Subsequent paragraphs are indented to show that they belong to the previous footnote.

```
{ some.code }
```

The whole paragraph can be indented, or just the first line. In this way, multi-paragraph footnotes work like multi-paragraph list items.

5 References

Allaire J, Xie Y, Dervieux C, et al. (2025) *Rmarkdown: Dynamic Documents for r*. Available at: https://github.com/rstudio/rmarkdown.

Hintz D and Otto S (2022) *GHformats: Templates for HTML and PDF/LaTeX Output Formats Designed for Guardant Health*. Available at: https://github.com/gh-dhintz/GHformats.

Kamm J (2000) *Evaluation of the Sedov-von Neumann-Taylor blast wave solution*. Technical Report LA-UR-00-6055. Los Alamos National Laboratory.

Knupp P (1999) Winslow smoothing on two-dimensional unstructured meshes. *Eng Comput* 15: 263–268.

R Core Team (2025) *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. Available at: https://www.R-project.org/.

Taylor G and Green A (1937) Mechanism of the production of small eddies from large ones. *P Roy Soc Lond A Mat* 158(895): 499–521.

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Acknowledgments

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